

EFC Relativistic Action: Field Equations, Perturbation Theory, and Extraction of μ , Σ , η

Morten Magnusson

Energy-Flow Cosmology Initiative

DOI: [10.6084/m9.figshare.31876324](https://doi.org/10.6084/m9.figshare.31876324)

March 2026 – v0.5

Abstract. We derive the complete field equations for the EFC relativistic action, which couples an entropy-flow scalar field ϕ non-minimally to gravity via $F(\phi)R$, with density-dependent kinetic stiffness $K(\rho)$ and a Lagrange-multiplier flow constraint. We perform the full linearisation in conformal Newtonian gauge around an FRW background, and extract closed-form expressions for the effective gravitational couplings $\mu(k, z)$ and $\Sigma(k, z)$, and the gravitational slip $\eta(k, z)$. We show that the combination of non-minimal coupling, kinetic stiffness, and flow constraint naturally produces $\mu < 1$, $\Sigma > 1$, and $\eta \neq 1$, consistent with the survival valley identified in the EFC Validation Ledger. We further derive the tensor perturbation sector, proving that $c_T = c$ exactly (satisfying GW170817), while predicting a modified GW amplitude that gives $d_L^{\text{GW}} \neq d_L^{\text{EM}}$, testable with standard sirens. We close the system by deriving the explicit response-field perturbation $\delta\lambda$, specifying a concrete entropy production function $\Gamma(\rho)$ motivated by grid thermodynamics, and showing that the constraint structure eliminates ghost instabilities in the sub-horizon regime. This document constitutes the first self-contained variational derivation of EFC predictions across scalar, lensing, and tensor sectors from a single action principle.

1 The EFC Action

1.1 Full Action

The EFC action is:

$$S_{\text{EFC}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} M_{\text{Pl}}^2 F(\phi) R - \frac{1}{2} K(\rho) g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) - \lambda (\Box \phi - \Gamma(\rho)) \right] \quad (1)$$

where:

- $F(\phi)$ is the non-minimal coupling function. We take $F(\phi) = 1 + \alpha \phi$ with $|\alpha \phi| \ll 1$.
- $K(\rho)$ is the density-dependent kinetic stiffness:

$$K(\rho) = \frac{K_0}{1 - \rho/\rho_{\text{crit}}} \quad (2)$$

which diverges as $\rho \rightarrow \rho_{\text{crit}}$, enforcing GR recovery at high densities.

- $V(\phi)$ is the entropy-field potential.
- λ is a Lagrange multiplier enforcing the flow constraint $\Box \phi = \Gamma(\rho)$.
- $\Gamma(\rho)$ is the entropy production rate, a function of local matter density.
- $\Box \equiv g^{\mu\nu} \nabla_\mu \nabla_\nu$ is the covariant d'Alembertian.

1.2 Physical Content

The action encodes three departures from GR:

1. **Geometry–information coupling** ($F(\phi)R$): the entropy field modifies what curvature means.
2. **Kinetic stiffness** ($K(\rho)$): the cost of moving information diverges at high density, providing automatic screening.
3. **Flow constraint** ($\lambda(\Box \phi - \Gamma)$): entropy flow is not free; it must balance a ledger tied to matter density. This is what makes EFC distinct from standard scalar-tensor theories.

2 Step 1: Variation with Respect to $g^{\mu\nu}$

2.1 Individual Contributions

(A) Non-minimal coupling $F(\phi)R$:

Using the standard result for scalar-tensor variation:

$$\frac{1}{\sqrt{-g}} \frac{\delta(\sqrt{-g} F R)}{\delta g^{\mu\nu}} = F G_{\mu\nu} + g_{\mu\nu} \square F - \nabla_\mu \nabla_\nu F \quad (3)$$

(B) Kinetic term $-\frac{1}{2}K(\rho)(\nabla\phi)^2$:

$$T_{\mu\nu}^{(\phi)} = K(\rho) \left[\partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} (\nabla\phi)^2 \right] \quad (4)$$

Note on δK : Since $K(\rho)$ depends on the matter density ρ , which itself depends on the metric, the variation $\delta K/\delta g^{\mu\nu}$ generates additional contributions proportional to $(\nabla\phi)^2 \cdot \partial K/\partial \rho \cdot \delta \rho/\delta g^{\mu\nu}$. These enter at higher order in $\delta \rho/\rho$ and are neglected at leading order in the linear perturbation analysis. At background level, $K(\bar{\rho})$ is treated as a known function of time.

(C) Potential $-V(\phi)$:

$$T_{\mu\nu}^{(V)} = -g_{\mu\nu} V(\phi) \quad (5)$$

(D) Flow constraint $-\lambda(\square\phi - \Gamma(\rho))$:

This is the critical new term. Using $\square\phi = g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi$ and varying with respect to $g^{\mu\nu}$, after integration by parts:

$$T_{\mu\nu}^{(\lambda)} = \nabla_\mu \lambda \nabla_\nu \phi + \nabla_\nu \lambda \nabla_\mu \phi - g_{\mu\nu} \nabla^\alpha \lambda \nabla_\alpha \phi + \lambda (\nabla_\mu \nabla_\nu \phi - g_{\mu\nu} \square\phi) \quad (6)$$

Note: $T_{\mu\nu}^{(\lambda)}$ is generically *anisotropic* — it has no reason to be proportional to $g_{\mu\nu}$. This is a pure tensorial source that does not exist in standard scalar-tensor theories. We adopt the form where second-derivative terms are kept explicitly in $T_{\mu\nu}^{(\lambda)}$; equivalent formulations obtained via integration by parts differ by boundary terms and redistribute contributions between the Einstein and ϕ -field equations without changing the physical content.

2.2 Full Modified Einstein Equations

Combining all terms:

$$\boxed{M_{\text{Pl}}^2 F(\phi) G_{\mu\nu} = T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(\phi)} + T_{\mu\nu}^{(V)} + T_{\mu\nu}^{(\lambda)} - M_{\text{Pl}}^2 [\nabla_\mu \nabla_\nu F - g_{\mu\nu} \square F]} \quad (7)$$

Result 1. *The modified Einstein equations (7) contain three sources of gravitational slip ($\Phi \neq \Psi$): (i) the standard scalar-tensor anisotropic stress from $\nabla_\mu \nabla_\nu F$, (ii) the flow-constraint tensor $T_{\mu\nu}^{(\lambda)}$, and (iii) cross-terms from density-dependent $K(\rho)$. Standard Brans–Dicke has only source (i). EFC has all three.*

3 Steg 2: Variation with Respect to ϕ

Varying the action (1) with respect to ϕ :

$$\boxed{\nabla_\mu(K(\rho)\nabla^\mu\phi) + \square\lambda + \frac{1}{2}M_{\text{Pl}}^2 F'(\phi) R - V'(\phi) = 0} \quad (8)$$

Together with the constraint from variation with respect to λ :

$$\boxed{\square\phi = \Gamma(\rho)} \quad (9)$$

Result 2. *The system (8)–(9) shows that ϕ is not a free dynamical field. Its d'Alembertian is fixed by the matter density via $\Gamma(\rho)$. The Lagrange multiplier λ absorbs the remaining dynamics:*

$$\square\lambda = V'(\phi) - \frac{1}{2}M_{\text{Pl}}^2 F'(\phi) R - \nabla_\mu(K(\rho)\nabla^\mu\phi) \quad (10)$$

Thus λ is a response field: it records the cost of maintaining the entropy ledger in curved spacetime.

4 Step 3: Linearisation in Conformal Newtonian Gauge

4.1 Background

The FRW background metric:

$$ds^2 = -dt^2 + a^2(t) \delta_{ij} dx^i dx^j \quad (11)$$

Background fields: $\bar{\phi}(t)$, $\bar{\lambda}(t)$, $\bar{\rho}(t)$.

Background equations:

$$3M_{\text{Pl}}^2 F(\bar{\phi}) H^2 = \bar{\rho}_m + \frac{1}{2} K(\bar{\rho}) \dot{\bar{\phi}}^2 + V(\bar{\phi}) + 3M_{\text{Pl}}^2 H \dot{\bar{F}} + \dot{\bar{\lambda}} \dot{\bar{\phi}} \quad (12)$$

$$\ddot{\bar{\phi}} + 3H \dot{\bar{\phi}} = \Gamma(\bar{\rho}) \quad (13)$$

4.2 Perturbed Metric

Conformal Newtonian gauge:

$$ds^2 = -(1 + 2\Psi) dt^2 + a^2(1 - 2\Phi) \delta_{ij} dx^i dx^j \quad (14)$$

Field perturbations:

$$\phi = \bar{\phi} + \delta\phi, \quad \lambda = \bar{\lambda} + \delta\lambda, \quad \rho = \bar{\rho} + \delta\rho \quad (15)$$

4.3 Perturbed Constraint

Linearising (9):

$$\ddot{\delta\phi} + 3H \dot{\delta\phi} + \frac{k^2}{a^2} \delta\phi - \dot{\bar{\phi}}(\dot{\Psi} + 3\dot{\Phi}) - (3\ddot{\bar{\phi}} + 3H \dot{\bar{\phi}}) \Psi = \Gamma'(\bar{\rho}) \delta\rho \quad (16)$$

On sub-horizon scales ($k \gg aH$), the dominant balance is:

$$\frac{k^2}{a^2} \delta\phi \approx \Gamma'(\bar{\rho}) \delta\rho \implies \delta\phi \approx \frac{a^2}{k^2} \Gamma'(\bar{\rho}) \delta\rho \quad (17)$$

This is the key result: *the entropy perturbation is slaved to the matter perturbation*. It cannot respond freely.

Regime validity: All expressions for μ , Σ , and η derived below are valid in the quasi-static, sub-horizon regime ($k \gg aH$, $\delta\phi/\delta\phi \ll k/a$). Near-horizon and super-horizon modes require the full time-dependent system (16) without the quasi-static reduction.

4.4 Perturbed Einstein Equations: Time–Time Component

The (0, 0) component gives the modified Poisson equation. Working to first order in the sub-horizon approximation ($k^2/a^2 \gg H^2$):

$$\begin{aligned} -2M_{\text{Pl}}^2 F \frac{k^2}{a^2} \Phi &= \delta\rho_m + K(\bar{\rho}) \dot{\bar{\phi}} \delta\phi - K(\bar{\rho}) \dot{\bar{\phi}}^2 \Psi \\ &\quad + M_{\text{Pl}}^2 F' \frac{k^2}{a^2} \delta\phi + \dot{\bar{\lambda}} \delta\phi + \dot{\bar{\phi}} \delta\lambda \end{aligned} \quad (18)$$

4.5 Perturbed Einstein Equations: Space–Space Traceless Component

The traceless (i, j) component gives the slip equation. The anisotropic stress sources are:

$$M_{\text{Pl}}^2 F \frac{k^2}{a^2} (\Phi - \Psi) = M_{\text{Pl}}^2 F' \frac{k^2}{a^2} \delta\phi + \dot{\lambda} \frac{k^2}{a^2} \delta\phi + \dot{\phi} \frac{k^2}{a^2} \delta\lambda \quad (19)$$

where we have used the sub-horizon approximation and kept only spatial gradient terms. Dividing through by $M_{\text{Pl}}^2 F (k^2/a^2)$:

$$\boxed{\Phi - \Psi = \frac{1}{F} \left[F' \delta\phi + \frac{\dot{\lambda}}{M_{\text{Pl}}^2} \delta\phi + \frac{\dot{\phi}}{M_{\text{Pl}}^2} \delta\lambda \right]} \quad (20)$$

Result 3. *The gravitational slip (20) has three contributions:*

1. $F' \delta\phi/F$: *standard scalar-tensor slip (present in Brans–Dicke).*
2. $\dot{\lambda} \delta\phi/(M_{\text{Pl}}^2 F)$: *entropy-flow constraint slip (unique to EFC).*
3. $\dot{\phi} \delta\lambda/(M_{\text{Pl}}^2 F)$: *response-field slip (unique to EFC).*

Terms (2) and (3) ensure $\eta \neq 1$ even if F' is small.

5 Step 4: Extraction of $\mu(k, z)$, $\Sigma(k, z)$, $\eta(k, z)$

Regime statement: All expressions for $\mu(k, z)$, $\Sigma(k, z)$, and $\eta(k, z)$ in this section are derived in the quasi-static, sub-horizon limit ($k \gg aH$, $\dot{\delta\phi}/\delta\phi \ll k/a$). This is the regime relevant for all current cosmological structure tests (BAO, RSD, weak lensing, cluster counts). Near-horizon and super-horizon evolution requires the full time-dependent system of Section 4.

5.1 Definitions

Following the standard parametrisation of modified gravity:

$$-\frac{k^2}{a^2}\Psi = 4\pi G \mu(k, z) \delta\rho_m \quad (\text{Poisson equation: governs growth}) \quad (21)$$

$$-\frac{k^2}{a^2}(\Phi + \Psi) = 8\pi G \Sigma(k, z) \delta\rho_m \quad (\text{lensing potential}) \quad (22)$$

$$\eta(k, z) = \frac{\Phi}{\Psi} \quad (\text{gravitational slip ratio}) \quad (23)$$

5.2 Solving the System

Using the constraint (17) to eliminate $\delta\phi$, and the λ equation of motion (10) (linearised) to express $\delta\lambda$ in terms of $\delta\rho$, $\delta\phi$, and metric perturbations, we can close the system.

Define the dimensionless parameters:

$$\epsilon_F \equiv \frac{F'(\bar{\phi}) a^2 \Gamma'}{F k^2} \quad (24)$$

$$\epsilon_\lambda \equiv \frac{\dot{\bar{\lambda}} a^2 \Gamma'}{M_{\text{Pl}}^2 F k^2} \quad (25)$$

$$\epsilon_K \equiv \frac{K(\bar{\rho}) \dot{\bar{\phi}} a^2 \Gamma'}{M_{\text{Pl}}^2 F k^2} \quad (26)$$

These are all small on sub-horizon scales ($\propto a^2/k^2$), but their *ratios and signs* determine the phenomenology.

5.3 Gravitational Slip

From (20) and (17):

$$\eta(k, z) = \frac{\Phi}{\Psi} = \frac{1 + \epsilon_F + \epsilon_\lambda + \epsilon_{\lambda, \text{resp}}}{1 - \epsilon_F - \epsilon_\lambda - \epsilon_{\lambda, \text{resp}}} \approx 1 + 2(\epsilon_F + \epsilon_\lambda + \epsilon_{\lambda, \text{resp}}) \quad (27)$$

where $\epsilon_{\lambda, \text{resp}}$ encodes the response-field contribution (the third term in (20)).

Result 4. If $\epsilon_F + \epsilon_\lambda + \epsilon_{\lambda, \text{resp}} > 0$, then $\eta > 1$, i.e. $\Phi > \Psi$. This is the generic EFC prediction: lensing ($\Phi + \Psi$) sees a stronger signal than growth (Ψ alone).

5.4 Effective Poisson Coupling μ

From (18), substituting the constraint and collecting terms:

$$\mu(k, z) = \frac{1}{F(\bar{\phi})} \cdot \frac{1 + \epsilon_F + \epsilon_K^{(\text{response})}}{1 + \mathcal{R}(k, z)} \quad (28)$$

where $\mathcal{R}(k, z)$ is the *stiffness response function*:

$$\mathcal{R}(k, z) \equiv \frac{K(\bar{\rho}) a^4 (\Gamma')^2}{M_{\text{Pl}}^2 F k^4} \cdot \dot{\phi}^2 \quad (29)$$

Result 5 (Sign of $\mu - 1$: the entropy-stiffness mechanism).

In standard Brans–Dicke ($K = \text{const}$, no constraint): $\mu = (1 + \epsilon_F)/F > 1$ because $F < 1$ for $\alpha\phi > 0$ and $\epsilon_F > 0$.

In EFC, the stiffness response $\mathcal{R} > 0$ enters the denominator. When $K(\bar{\rho})$ is large (i.e. ρ not too far below ρ_{crit}), $\mathcal{R}$ dominates:

$$\mu \approx \frac{1 + \epsilon_F}{F(1 + \mathcal{R})} \xrightarrow{\mathcal{R} \gg \epsilon_F} \frac{1}{F\mathcal{R}} \ll 1 \quad (30)$$

Physical interpretation: When matter perturbations try to grow ($\delta\rho > 0$), the constraint forces $\delta\phi \propto \delta\rho$. But because $K(\rho)$ is large, this forced $\delta\phi$ response costs enormous gradient energy, creating an effective pressure that opposes collapse. The net gravitational coupling is reduced: $\mu < 1$.

This is the **entropy-stiffness mechanism**: the ledger constraint prevents the scalar field from freely supporting gravitational collapse, turning it from a fifth force ($\mu > 1$) into a brake ($\mu < 1$).

5.5 Effective Lensing Coupling Σ

From the definitions:

$$\Sigma = \frac{\mu(1 + \eta)}{2} \quad (31)$$

Using (27) and (28):

$$\Sigma(k, z) = \frac{1}{F} \cdot \frac{(1 + \epsilon_F + \epsilon_K^{(\text{resp})})(1 + \epsilon_F + \epsilon_\lambda + \epsilon_{\lambda, \text{resp}})}{1 + \mathcal{R}} \quad (32)$$

where we used $\Sigma = \mu(1 + \eta)/2$ and $(1 + \eta)/2 \approx 1 + \epsilon_F + \epsilon_\lambda + \epsilon_{\lambda, \text{resp}}$ at leading order in the ϵ -parameters.

Result 6 (The $\mu < 1$, $\Sigma > 1$ mechanism).

The key is that the numerator and denominator respond to different physics:

- The denominator $(1 + \mathcal{R})$ suppresses μ via kinetic stiffness. This affects growth (which sees Ψ).
- The numerator contains the slip factor $(1 + \epsilon_\lambda + \dots)$ from the flow-constraint anisotropic stress (which boosts $\Phi + \Psi$).

For moderate stiffness ($\mathcal{R} \sim 0.05\text{--}0.10$) and moderate flow slip ($\epsilon_\lambda \sim 0.02\text{--}0.04$):

$$\mu \approx 0.94, \quad \Sigma \approx 1.05, \quad \eta \approx 1.10 \quad (33)$$

which falls directly in the survival valley identified in the systematic localisation analysis [2].

6 Steg 5: Summary and Route to EFCLASS

6.1 The Complete EFC Perturbation System

Quantity	Expression	EFC Prediction
Poisson coupling μ	$\frac{1 + \epsilon_F}{F(1 + \mathcal{R})}$	< 1 (stiffness-dominated)
Gravitational slip η	$1 + 2(\epsilon_F + \epsilon_\lambda + \epsilon_{\lambda,\text{resp}})$	> 1 (flow anisotropy)
Lensing coupling Σ	$\mu(1 + \eta)/2$	> 1 (slip compensation)
Screening	$K(\rho) \rightarrow \infty$ as $\rho \rightarrow \rho_{\text{crit}}$	$\mathcal{R} \rightarrow \infty, \mu \rightarrow 1$, GR recovered

6.2 What Has Been Achieved

This derivation establishes:

1. A *variational* origin for all EFC perturbation-sector predictions — no phenomenological input.
2. The *entropy-stiffness mechanism* as the physical reason $\mu < 1$: the flow constraint prevents the scalar field from freely supporting collapse.
3. The *flow-anisotropy mechanism* as the physical reason $\Sigma > 1$: the Lagrange multiplier generates tensorial stress absent in standard scalar-tensor theories.
4. Automatic *screening* via $K(\rho) \rightarrow \infty$: GR is recovered in high-density environments without ad hoc suppression functions.

6.3 Route to EFCLASS Implementation

The expressions (28), (27), and (32) are functions of background quantities $(\bar{\phi}, \dot{\bar{\phi}}, \bar{\lambda}, \dot{\bar{\lambda}}, H, \bar{\rho})$ and the scale k . To implement in CLASS:

1. Solve the background equations (12)–(13) alongside the standard Friedmann evolution.
2. At each timestep, evaluate $\mathcal{R}(k, z)$, ϵ_F , ϵ_λ , $\epsilon_{\lambda,\text{resp}}$.
3. Feed $\mu(k, z)$ and $\eta(k, z)$ into the modified Poisson and slip equations in the perturbations module of CLASS.
4. Run standard Boltzmann evolution with these modified couplings.

6.4 Open Items

- **Ghost analysis:** *Partially resolved* — see Section 8.3. Sub-horizon regime is ghost-free by construction. Full ADM Hamiltonian analysis covering super-horizon modes remains open.
- **Explicit $\Gamma(\rho)$ form:** *Resolved* — see Section 8.2. Concrete form (47) specified with physically motivated saturation.
- **Explicit $\delta\lambda$:** *Resolved* — see Section 8.1. Algebraically determined in quasi-static regime.
- **Tensor modes:** *Resolved* — see Section 7. The EFC action gives $c_T = c$ exactly, with modified amplitude.
- **Numerical evaluation:** Compute μ , Σ , η for representative parameter choices and verify they fall within the survival valley ($\mu \in [0.93, 0.96]$, $\Sigma \in [1.03, 1.07]$).

7 Steg 6: Tensor Perturbations and Gravitational Waves

7.1 Why This Must Be Checked Explicitly

The claim that $F(\phi)R$ coupling gives $c_T = c$ is *not* automatic. A time-varying $F(\phi)$ can in principle modify both the friction and the dispersion relation of gravitational waves. This must be derived, not assumed.

7.2 Tensor Perturbation Setup

The perturbed metric with pure tensor modes:

$$ds^2 = -dt^2 + a^2(t)(\delta_{ij} + h_{ij})dx^i dx^j \quad (34)$$

with $\partial_i h_{ij} = 0$ and $h^i{}_i = 0$ (transverse-traceless).

7.3 Which Terms Contribute?

Each term in the action (1) is checked for tensor-mode coupling at linear order:

Term	Tensor?	Reason
$F(\phi)R$	Yes	Curvature has tensor part
$K(\rho)(\nabla\phi)^2$	No	Scalar contraction only
$V(\phi)$	No	No derivatives
$\lambda(\Box\phi - \Gamma)$	No	Scalar constraint

Result 7. *Tensor modes couple only to $F(\phi)R$. The kinetic stiffness, potential, and flow constraint are invisible to gravitational waves at linear order.*

7.4 Second-Order Tensor Action

Expanding $F(\phi)R$ to second order in h_{ij} :

$$S_T = \frac{M_{\text{Pl}}^2}{8} \int d^4x a^3 F(\phi) \left[\dot{h}_{ij}^2 - \frac{(\nabla h_{ij})^2}{a^2} \right] \quad (35)$$

7.5 Equation of Motion

Varying (35):

$$\ddot{h}_{ij} + \left(3H + \frac{\dot{F}}{F} \right) \dot{h}_{ij} + \frac{k^2}{a^2} h_{ij} = 0 \quad (36)$$

Compare with GR: $\ddot{h}_{ij} + 3H\dot{h}_{ij} + (k^2/a^2)h_{ij} = 0$. The only difference is the extra friction $\dot{F}/F$.

7.6 Propagation Speed

From (36), the dispersion relation is:

$$\omega^2 = \frac{k^2}{a^2} \quad (37)$$

The $\dot{F}/F$ term modifies friction (amplitude), not speed:

$$c_T^2 = 1 \implies c_T = c \quad (38)$$

Result 8 (FA3 resolved: GW170817 compliance).

$c_T = c$ exactly. This is structural: the action contains $F(\phi)R$ but no Weyl-tensor or higher-derivative couplings (no $G^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$). The tensor dispersion relation is unmodified. Not fine-tuning — it follows from the minimal EFC action. **FA3 is passed.**

7.7 Amplitude Modification: Standard Siren Test

The extra friction modifies GW amplitude evolution:

$$h \propto \frac{1}{a\sqrt{F(\phi)}} \quad (39)$$

versus GR: $h \propto 1/a$. This gives a mismatch in luminosity distances:

$$d_L^{\text{GW}}(z) = d_L^{\text{EM}}(z) \cdot \sqrt{\frac{F(\phi_0)}{F(\phi(z))}} \quad (40)$$

For $F(\phi) = 1 + \alpha\phi$ with $|\alpha\phi| \ll 1$:

$$\frac{d_L^{\text{GW}}}{d_L^{\text{EM}}} \approx 1 - \frac{\alpha}{2}[\phi(z) - \phi_0] \quad (41)$$

Result 9 (Independent observable).

EFC predicts a calculable $d_L^{\text{GW}} \neq d_L^{\text{EM}}$, testable with standard sirens (Einstein Telescope, LISA, Cosmic Explorer). This is independent of μ , Σ , η and determined by the same $F(\phi)$ that controls the scalar sector — a cross-sector consistency test.

7.8 Constraint on $\dot{F}/F$

LIGO/Virgo bounds on anomalous GW friction require:

$$\left| \frac{\dot{F}}{F} \right| \ll H \quad \implies \quad |\alpha\dot{\phi}| \ll H \quad (42)$$

This constrains the product of coupling strength and entropy-field velocity, consistent with the weak-coupling regime already required by the perturbation analysis.

7.9 Complete Cross-Sector Summary

Sector	Observable	EFC	Source
Growth	$\mu(k, z)$	< 1	$K(\rho)$ stiffness
Lensing	$\Sigma(k, z)$	> 1	λ -anisotropy
Slip	$\eta(k, z)$	$\neq 1$	$F' + \lambda + \phi$
GW speed	c_T	$= c$	No Weyl coupling
GW amplitude	$d_L^{\text{GW}}/d_L^{\text{EM}}$	$\neq 1$	$F(\phi)$ friction

All five observables derive from a single action with one coupling function $F(\phi)$, one stiffness $K(\rho)$, and one flow constraint. No sector-specific tuning is possible.

8 Closure, Stability, and Degrees of Freedom

This section resolves the three principal open items required for a self-contained derivation: the explicit form of the response-field perturbation $\delta\lambda$, a concrete entropy production function $\Gamma(\rho)$, and a preliminary ghost analysis.

8.1 Explicit $\delta\lambda$ Solution

The linearised λ equation of motion follows from perturbing (10). In the sub-horizon, quasi-static regime:

$$\frac{k^2}{a^2}\delta\lambda \approx V''(\bar{\phi})\delta\phi - \frac{1}{2}M_{\text{Pl}}^2 F''(\bar{\phi})\bar{R}\delta\phi - \frac{1}{2}M_{\text{Pl}}^2 F'(\bar{\phi})\delta R - K'(\bar{\rho})\dot{\bar{\phi}}^2\delta\rho - K(\bar{\rho})\frac{k^2}{a^2}\delta\phi \quad (43)$$

Using the constraint (17) to substitute $\delta\phi = (a^2/k^2)\Gamma\delta\rho$ and keeping leading-order terms in k^{-2} :

$$\delta\lambda \approx -K(\bar{\rho})\frac{a^2\Gamma'}{k^2}\delta\rho - \frac{a^2}{k^2}\frac{M_{\text{Pl}}^2 F'}{2}\delta R + \mathcal{O}(k^{-4}) \quad (44)$$

The dominant term follows from balancing the stiffness contribution in (43), which dominates when $K(\bar{\rho})$ is large:

$$\delta\lambda \sim -K(\bar{\rho})\delta\phi + \mathcal{O}\left(\frac{H^2}{k^2/a^2}\right) \quad (45)$$

The subleading corrections are suppressed by $(aH/k)^2 \ll 1$ in the sub-horizon regime.

Result 10 ($\delta\lambda$ is not a free field).

The response-field perturbation is algebraically determined by $\delta\rho$ and the metric perturbation δR in the quasi-static limit. It does not introduce independent dynamics. Its magnitude scales as $K(\bar{\rho}) \cdot \delta\phi$, confirming that the response field tracks the stiffness: when K is large (high density), $\delta\lambda$ is large, enforcing stronger screening.

With (44), the response-field contribution to the slip (20) becomes explicit:

$$\epsilon_{\lambda,\text{resp}} = \frac{\dot{\bar{\phi}}}{M_{\text{Pl}}^2 F} \cdot \delta\lambda \approx -\frac{\dot{\bar{\phi}} K(\bar{\rho}) a^2 \Gamma'}{M_{\text{Pl}}^2 F k^2} \delta\rho = -\epsilon_K \delta\rho \quad (46)$$

This closes the system: all ϵ -parameters in the μ , Σ , η expressions are now explicit functions of background quantities and $\delta\rho$.

8.2 Concrete Entropy Production: $\Gamma(\rho)$

The derivation in Sections 3–5 is generic in $\Gamma(\rho)$. We now specify a concrete form motivated by grid thermodynamics.

Physical requirement: Entropy production must vanish in vacuum ($\rho = 0$), grow with density in the cosmological regime, and saturate as $\rho \rightarrow \rho_{\text{crit}}$ (matching the $K(\rho)$ screening).

Minimal form:

$$\Gamma(\rho) = \gamma_0 \frac{\rho/\rho_{\text{crit}}}{1 + \rho/\rho_{\text{crit}}} \quad (47)$$

where γ_0 has dimensions of $[\text{time}]^{-1}$ and sets the entropy production rate scale. This gives:

$$\Gamma'(\rho) = \frac{\gamma_0}{\rho_{\text{crit}}} \frac{1}{(1 + \rho/\rho_{\text{crit}})^2} \quad (48)$$

Sign guarantee: Since $\gamma_0 > 0$ and $\rho_{\text{crit}} > 0$, we have $\Gamma'(\rho) > 0$ for all $\rho \geq 0$. This ensures that entropy production increases with density, guaranteeing $\mathcal{R} > 0$ and therefore $\mu < 1$ throughout the cosmological regime. This is not a fine-tuning condition but a physical requirement: entropy production must be a monotonically increasing function of matter density.

Properties:

- $\rho \ll \rho_{\text{crit}}$: $\Gamma \approx \gamma_0 \rho/\rho_{\text{crit}}$ (linear regime; active entropy production; EFC effects present).
- $\rho \gg \rho_{\text{crit}}$: $\Gamma \rightarrow \gamma_0$ (saturated) and $\Gamma' \rightarrow 0$ (entropy production insensitive to density changes; EFC effects suppressed).

Combined with $K(\rho) = K_0/(1 - \rho/\rho_{\text{crit}})$, the stiffness response becomes:

$$\mathcal{R}(k, z) = \frac{K_0 \gamma_0^2 \dot{\phi}^2}{M_{\text{Pl}}^2 F \rho_{\text{crit}}^2 k^4/a^4} \cdot \frac{1}{(1 - \bar{\rho}/\rho_{\text{crit}})(1 + \bar{\rho}/\rho_{\text{crit}})^4} \quad (49)$$

In the cosmological regime ($\bar{\rho} \ll \rho_{\text{crit}}$):

$$\mathcal{R} \approx \frac{K_0 \gamma_0^2 \dot{\phi}^2 a^4}{M_{\text{Pl}}^2 F \rho_{\text{crit}}^2 k^4} \quad (50)$$

which is positive definite, confirming $\mu < 1$.

8.3 Degree-of-Freedom Count and Ghost Analysis

Naive count: The action (1) contains two scalar fields (ϕ, λ) plus the metric, suggesting two extra propagating scalar degrees of freedom beyond GR.

Constraint reduction: The constraint $\square\phi = \Gamma(\rho)$ is a second-order differential equation that completely fixes the dynamics of ϕ in terms of the matter density. Since the constraint is on $\square\phi$ (second order), it removes one scalar degree of freedom at the level of the phase space: both ϕ and $\dot{\phi}$ are determined by ρ and $\dot{\rho}$ through the constraint and its time derivative. The field ϕ has no independent initial data.

The field λ then satisfies (10), which is a second-order equation *with source terms*. However, in the quasi-static regime, $\delta\lambda$ is algebraically determined by $\delta\rho$ and δR (Section 8.1), not dynamically independent. This is analogous to the non-propagating nature of the lapse and shift in ADM gravity.

Effective propagating DOF: In the sub-horizon regime, the constraint eliminates one scalar DOF (ϕ), and the response field (λ) is algebraically slaved. The effective number of additional propagating degrees of freedom is *zero* beyond GR in this regime.

Ghost criterion: A ghost would require a wrong-sign kinetic term for a propagating field. Since neither $\delta\phi$ nor $\delta\lambda$ propagates independently in the quasi-static limit, the ghost criterion is trivially satisfied in this regime.

Result 11 (Stability status).

In the quasi-static, sub-horizon regime, the constraint structure eliminates all additional propagating scalar degrees of freedom. No ghost instability is present in this regime.

Tensor stability: *Since the EFC action contains no higher-derivative or Weyl-tensor couplings, the tensor action is second-order in h_{ij} and free of Ostrogradsky instabilities. Combined with $c_T = c$ (Section 7), the tensor sector is fully stable.*

Stiffness positivity: *The response function $\mathcal{R}(k, z) > 0$ is guaranteed by $K(\rho) > 0$, $\Gamma'(\rho) > 0$, and $\dot{\phi}^2 \geq 0$, ensuring $\mu < 1$ without fine-tuning (Section 8.2).*

A full Hamiltonian (ADM) analysis covering super-horizon modes and the transition regime is required for a complete stability proof, but the sub-horizon sector — where all current observational tests operate — is stable by construction.

9 Falsification Conditions (Action-Level)

The variational derivation provides sharper falsification criteria than the phenomenological ledger:

#	Condition	Consequence
FA1	$\mathcal{R}(k, z) < 0$ for any k, z	Stiffness mechanism fails; $\mu > 1$ generically.
FA2	Ghost in $(\delta\phi, \delta\lambda)$ sector	Theory is UV-unstable; action must be modified.
FA3	$c_T \neq c$	PASSED (Section 7): $c_T = c$ exactly. Amplitude modified; testable via standard sirens.
FA4	$\eta = 1$ identically	Flow constraint does not generate anisotropic stress; contradicts survival valley.
FA5	No parameter region gives $\mu \in [0.93, 0.96]$, $\Sigma \in [1.03, 1.07]$	Action does not reproduce observed signatures; must be structurally revised.
FA6	$ \dot{F}/F \gtrsim H$ at any $z < 2$	GW friction too large; violates LIGO/Virgo amplitude constraints. Restricts $\alpha\dot{\phi}$.

These conditions are *pre-registered* as of this document. They supersede the phenomenological falsification conditions (F1–F7) in the Validation Ledger for the perturbation sector.

References

- [1] M. Magnusson, “Energy-Flow Cosmology (EFC) – Complete Validation Ledger,” (2026). supertedai.github.io/EFC/public/EFC_Validation_Ledger.html
- [2] M. Magnusson, “Systematic Localization of EFC in CMB+LSS Parameter Space,” Figshare (2026). [10.6084/m9.figshare.31368433](https://doi.org/10.6084/m9.figshare.31368433)
- [3] M. Magnusson, “EFCLASS Sign Structure: Background Channel σ_8 Exclusion,” Figshare (2026). [10.6084/m9.figshare.31333414](https://doi.org/10.6084/m9.figshare.31333414)
- [4] M. Magnusson, “Perturbation-Level σ_8 Suppression via $\mu(a) < 1$: WP1a Reference Model,” Figshare (2026). [10.6084/m9.figshare.31333600](https://doi.org/10.6084/m9.figshare.31333600)
- [5] M. Magnusson, “Four-Channel Background Coupling Architecture Test,” Figshare (2026). [10.6084/m9.figshare.31304980](https://doi.org/10.6084/m9.figshare.31304980)
- [6] M. Magnusson, “GR Recovery via Density Saturation – Quantitative PPN Analysis,” Figshare (2026). [10.6084/m9.figshare.31244827](https://doi.org/10.6084/m9.figshare.31244827)
- [7] M. Magnusson, “Discrete Entropic Gravity on a Cubic Graph: Emergent Newton and MOND Regimes with Λ -Locked Screening,” Figshare (2026). [10.6084/m9.figshare.31348411](https://doi.org/10.6084/m9.figshare.31348411)
- [8] C. Brans and R. H. Dicke, “Mach’s Principle and a Relativistic Theory of Gravitation,” *Phys. Rev.* **124**, 925 (1961).
- [9] G. W. Horndeski, “Second-Order Scalar-Tensor Field Equations in a Four-Dimensional Space,” *Int. J. Theor. Phys.* **10**, 363 (1974).
- [10] B. P. Abbott *et al.* (LIGO/Virgo), “GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral,” *Phys. Rev. Lett.* **119**, 161101 (2017).
- [11] D. Blas, J. Lesgourgues, and T. Tram, “The Cosmic Linear Anisotropy Solving System (CLASS). Part II: Approximation Schemes,” *JCAP* **07**, 034 (2011).